Research Project Technical Record

Parameter-Efficient Deepfake Detection: Complete Research Journey, Mathematical Foundations, Architecture, Implementation and Experimental Results

Generator-Balanced Latent Adaptation (GBLA) project

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Status of this document. Record of the research stage that ends on 5 November 2026 (repository commit `06aedba`). Every number in this report was read from the executed notebook outputs in the repository; items that could not be traced to a file are marked **unverified** and are not used as results. Nothing in the implementation, datasets or experiment configurations was changed while preparing this report.

5 November 2026

Executive Summary

Goal. Build a deepfake detector that (i) trains as few parameters as possible on top of a frozen pretrained model and (ii) generalises to manipulation methods ("generators") and datasets not seen in training. The project is framed as *parameter-efficient optimisation*: the question is how to train a very small set of parameters so that the result transfers, not how to compress a representation or a weight manifold.

What was built and run (M0–M9b, 5 Oct – 5 Nov 2026). A frozen OpenAI CLIP ViT-L/14 visual encoder (303,181,826 parameters in the detector) in which only the 50 LayerNorms and a linear head on the L2-normalised 1024-d CLS token are trained: **104,450 trainable parameters (0.0345%)**. On FaceForensics++ c23 this baseline (a reproduction of LNCLIP-DF [5]) reaches 98.04 seen-generator video AUC and 88.37 on the unseen FaceShifter generator, within 0.8 points of the published numbers on every test set. Two attempts to shrink the trainable set further by a seeded random-subspace latent ($\theta = \theta_0 + \alpha W z$, M2/M2b) **failed**: with $d \leq 1024$ the latent learned little beyond the class ratio or the head alone. A curvature-based explanation of generator dominance (M3) also **failed** its own checks, although a geometric measurement showed that pooled training is dominated by Face2Face and nearly ignores NeuralTextures.

The method that survived is **GBLA**: during training each generator g uses $\theta_s + \delta_g$ with a penalty $\lambda\|\delta_g\|^2$, batches are pure (one generator at a time) and only the shared θ_s (104,450 numbers) is deployed. With $\lambda = 0.01$:

- FaceShifter (unseen, FF++) video AUC **92.91** vs 90.53 for pooled training with the same batching and 88.19 for the LNCLIP-DF reproduction at the same 10-epoch budget;
- leave-one-generator-out mean held-out AUC **88.04** vs 87.49 pooled and 85.34 for a Fisher merge, but *worse* when NeuralTextures is held out (62.29 vs 64.40);
- cross-dataset DFD video AUC **96.71** [95.30, 98.21] vs 95.88 for the reproduction (paired difference +0.82 [+0.02, +1.73]), and UADFV **99.63** vs 99.25 (difference not significant; UADFV is near ceiling).

The parameter count of the deployed detector is unchanged from LN-tuning; GBLA changes how those parameters are optimised. All results are single-seed.

Reference values supplied for verification. The total of 303,966,208 parameters is verified, but it is the full ViT-L/14 visual tower *including* the unused 786,432-parameter text projection (M0). The value "786,432 trainable (0.2587%)", the LoRA adapters on `q_proj`/`v_proj`, the codebook utilisation (97/1024, 46/1024) and the accuracy/loss figures (98.38%, 97.56%, 0.134335, 97.125%) do not appear in any notebook, log, commit or document of this project; no LoRA or codebook code exists. They are reported as **unverified** (Section 9.10).

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1 Research Motivation and Problem Formulation

1.1 Why deepfake detection matters

Face-manipulation tools can replace identities (face swapping) or re-enact expressions in real videos. Detectors are needed wherever video is used as evidence or to inform the public. The practical difficulty is not detecting a known manipulation method; it is detecting a method that was not available when the detector was trained.

1.2 Limitations of conventional detectors

- **Generator-specific artifacts.** A detector trained on a few manipulation methods (generators) learns cues that are specific to them, for example blending seams or texture statistics. Same-generator accuracy on FaceForensics++ (FF++) [8] is close to saturation and hides this weakness.
- **Generalisation gaps.** Accuracy on an unseen generator, or on a different dataset with other capture conditions and compression, is usually much lower. Two kinds of shift must be kept apart: *cross-generator* (same dataset, new manipulation method) and *cross-dataset* (new data source).
- **Cost of fine-tuning large models.** Large pretrained backbones such as CLIP [7] transfer better than small CNNs, but fine-tuning all of their roughly 303 M parameters is expensive, needs much memory for optimiser state, and can overfit to the training generators.

1.3 Research objective

> *Develop a parameter-efficient deepfake detection framework that minimises the number of trainable parameters while retaining detection performance and improving generalisation across different deepfake generation methods.*

The intent is to keep the representational power of a frozen pretrained model and adapt only a very small set of parameters. Throughout this report, "parameter-efficient" means a small number of *trained and stored* parameters. The frozen backbone is still part of the model and is always reported (Total / Trainable / Frozen / Trainable%). Inference compute is not reduced by any method in this project.

1.4 How the objective evolved

1. **Literature phase (24–29 Sep 2026).** Five papers were analysed (Section 3). The common assumption that "fewer trainable parameters ⇒ better generalisation" was found to be contradicted by LNCLIP-DF's own LoRA result, and generator dominance in pooled training was identified as an unmeasured quantity.
2. **Initial proposal (29 Sep – 1 Oct).** "One Generator, One Vote": measure how unequally generators influence pooled training (effective number G_{eff}) and rebalance them, all inside a tiny seeded latent so that the deployed adaptation would be about 65 numbers. The method name GBLA (Generator-Balanced Latent Adaptation) dates from this stage.
3. **Compact latent abandoned (M2/M2b, 14–15 Oct).** The latent could not represent the LayerNorm adaptation at small d (Section 9.3). The compact-payload claim was dropped.
4. **Curvature explanation abandoned (M3, 17 Oct).** The Hessian-based weighting did not explain the pooled solution; only the geometric dominance measurement survived.
5. **Current method (M4 onward, 17 Oct – 5 Nov).** GBLA as shared-plus-private optimisation directly on the 104,450 LayerNorm and head parameters, evaluated by leave-one-generator-out (M5), against merging baselines (M6), and across datasets (M8b, M9b).

The mid-semester report and slides of 3 November 2026 still describe stage 2 (latent, 65 deployed numbers, "progress up to M2") although M3–M7 had been run by 20 October. That document is therefore out of date with respect to the method; this report supersedes it.

2 Sources, Investigation Method and Chronology

2.1 What exists in the workspace

The investigation covered the workspace `E:\NITR\3rd Semester\Satyabrata Research`, the git history of the repository, the paper folder, the mid-semester folder and the project conversation record.

Table 1: Inventory of project artefacts examined.

Artefact	Content and use in this report
Repository notebooks `M0.ipynb` … `M9b.ipynb`	14 executed notebooks (M0, M1, M2, M2b, M3, M4, M5, M6, M7, M7b, M8a, M8b, M9a, M9b), each holding code, configuration and outputs. **Primary source of every number.**
Git history (16 commits)	Dates and order of milestones (Table 2).
`unzip_dfd_c23.ipynb`, `unzip_celebdf_v2.ipynb`	Dataset download/unzip helpers outside the repository; not part of the research pipeline.
`Thesis Papers/` (8 PDFs)	Papers 4–8 analysed in the literature phase; papers 1–3 are background (Transformer, PCA tutorial, Transformer slides).
Mid-semester GBLA report (PDF) and two slide decks (`.pptx`)	Earlier GBLA documentation (state up to M2).
`Mid Semester/` `RP_MID_SEM/`	A separate multimodal (audio-visual) project report and proxy-task metrics; *not* part of this project. It is the only place where Tian et al. [1] is cited.
Project conversation record	Literature analysis, design decisions, bugs and their fixes.
Not present	No `.py` modules, no README, no training logs outside notebooks, no checkpoints in the repository (checkpoints stay on C-DAC under `runs/`), no LoRA or codebook code.

Verification policy. A number is reported as a result only if it appears in an executed notebook output. Where a quantity was planned but never measured, it is marked **planned** or *not recorded*. Where a value was supplied from earlier discussion but has no source file, it is marked **unverified**.

2.2 Chronology

Table 2: Chronology reconstructed from the conversation record and the git log.

Date (2026)	Commit	Activity	Status
24 Sep	–	Analysis of papers 4–8 (Section 3); hypotheses A–F and candidates C1–C5	**completed**
29 Sep – 1 Oct	–	GBLA proposal (“One Generator, One Vote”), pipeline design, claims audit	**completed**
5 Oct	`2572354`	Initial code (repository created)	**completed**
8 Oct	`83cd5dc`	M0 data pipeline: 192,000 paired face crops	**completed**
13 Oct	`3e62dfe`	M1 LNCLIP-DF reproduction (20 epochs)	**completed**
14 Oct	`dfc41db`	M2 latent re-parameterisation of LN + head, $d \in \{16, \ldots, 4096\}$	**failed**
15 Oct	`6549a0e`	M2b latent LN with directly trained head	**failed**
17 Oct	`9f1d12d`	M3 E1 go/no-go: per-generator solutions, curvature, G_{eff}	NO-GO (curvature)
18 Oct	`0ef0d1f`	M4 GBLA λ sweep	**completed**
19 Oct	`0731e63`	M5 leave-one-generator-out (LOGO) at $\lambda = 0.01$	**completed**
19 Oct	`04aa840`	M6 Fisher / uniform merge baselines	**completed**
20 Oct	`5ef28c8`, `7a6758d`	M7 G_{eff} law over 25 subsets; M7b re-analysis	**completed** (law not supported)
3 Nov	–	Mid-semester report and slides (describe state up to M2)	superseded
4 Nov	`439e77a`, `6661977`	M8a DFD preprocessing; M8b DFD cross-dataset test	**completed**
5 Nov	`f8f6bb7`, `06aedba`	M9a UADFV preprocessing; M9b UADFV cross-dataset test	**completed**

3 Literature Review

The literature phase started on 24 September 2026 with papers 4–8 of the thesis paper folder. Files 7 and 8 are two versions of the same work (LNCLIP-DF), so four distinct contributions were analysed. Tian et al. (Paper 1 in the task specification) is not part of that set; it appears only in a separate audio-visual project report stored

in the same mid-semester folder. It is reviewed here because it was requested, with its bibliographic details verified against the arXiv record. No "CVPR 2025 generalisable video-based deepfake detection" paper appears anywhere in the project record, so none is reviewed. Statements below about a paper's content are limited to what was read from the paper during the literature phase or, for Tian et al., from the arXiv text.

3.1 Paper 1 — Tian et al., unsupervised multimodal detection

Citation. M. Tian, M. Khayatkhoei, J. Mathai, W. AbdAlmageed, "Unsupervised Multimodal Deepfake Detection Using Intra- and Cross-Modal Inconsistencies", arXiv:2311.17088, 2023 [1]. A later version appeared at BMVC 2025 under the title "Unsupervised Multimodal Deepfake Detection Through Explicit Intra-Modal and Cross-Modal Inconsistency Discovery".

Problem and motivation. Supervised detectors need labelled fakes and do not generalise to new generation methods. The authors detect deepfake videos without fake training data by finding inconsistencies inside a single video.

Method. The central hypothesis is that motion-transfer deepfakes face a trade-off: if the motion of the source is transferred exactly, identity becomes inconsistent across frames; if identity is kept exact, motion is distorted and no longer matches the audio. They support this with an experiment in which identity is predicted from facial-landmark motion vectors, which works much better on real than on fake videos. Two scores are computed over time windows of a video: an *intra-modal* identity-consistency score (similarity of identity features of the same video across windows) and a *cross-modal* audio–video consistency score (similarity between audio and visual features of the same window). Networks are trained with consistency losses built from similarity matrices across identities and time windows, on real VoxCeleb2 videos only.

Results (as reported). Average AUC 96.81% on FakeAVCeleb, above prior unsupervised methods; trained only on VoxCeleb2 with FakeAVCeleb identities removed.

Strengths and limitations. No fake data in training, explainable (localises inconsistent windows), no per-identity reference video. It needs audio and talking-face video and is designed for motion-transfer fakes; it does not address single-frame face images.

Relation to this project. None of its components were adopted. GBLA is a visual-only, frame-level, supervised method. The idea of training without fake labels was considered in the literature phase through the audio paper CA-SOADD (candidate C3 below), not through this paper. Exact loss equations of this paper are not reproduced here because they were not part of the project analysis.

3.2 Paper 4 — Mapping Networks (Sen & Mukherjee)

Citation. L. Sen, S. Mukherjee, "Mapping Networks", CVPR 2026 [2] (NIT Rourkela).

Objective and method. Replace training of a high-dimensional weight vector by training a small latent $z \in \mathbb{R}^d$ that generates the weights through a fixed mapping: $\hat{\theta} = \sigma(Wz + b)$, $W \in \mathbb{R}^{P \times d}$, with a Mapping Loss $L_{\text{map}} = L_{\text{task}} + \lambda_{\text{st}} L_{\text{stab}} + \lambda_{\text{sm}} L_{\text{smooth}} + \lambda_{\text{al}} L_{\text{align}}$. Deepfake detection is one of five demonstration tasks.

Observations made in the project analysis. (i) Trainable parameters are tiny (e.g. 2,048), but the fixed mapping matrix must be stored: for a 537,994-parameter target network and $d = 2048$, $|W| \approx 1.10 \times 10^9$ entries (about 4.4 GB in fp32), which the paper does not report. (ii) With the stated modulation $w_{ij} \leftarrow w_{ij} + \alpha z_i$, the added term is $\alpha \|z\|^2$, the same for every output index, so "weight modulation" adds a single shared scalar. (iii) Deepfake results are same-dataset accuracies (Celeb-DF, FF++) with no cross-generator or cross-dataset test.

Adopted / not adopted. The *idea* of training a low-dimensional latent that generates parameters inspired the M2/M2b latent re-parameterisation, applied only to the 104,450 LayerNorm and head parameters and with an explicit offset θ_0 and a seeded (not stored) W. That experiment failed (Section 9.3). The non-linearity σ, the modulation and the Mapping Loss terms were not adopted. In the final framing Mapping Networks is related work, not the basis of the method.

3.3 Paper 5 — Breaking Manifold Continuity (Wang et al.)

Citation. C. Wang, J. Wang, K.-Y. Zhang, T. Yao, S. Ding, R. Yi, L. Ma, "Breaking Manifold Continuity: Vector Quantized Modeling for Real-Centric Deepfake Detection", ICML 2026 (PMLR) [3].

Method. A learnable codebook ($K = 1024$, $D = 1024$) quantises CLIP ViT-L/14 patch tokens; a 2-block masked ViT restores context; per-code tangent bases $U_k \in \mathbb{R}^{1024 \times 16}$ separate valid real variation from off-

manifold residuals; a hypernetwork conditioned on the CLS token fuses the residuals. Training uses FF++ c23 with 8 frames per video; testing uses 32 frames per video on six datasets and six DF40 methods.

Parameter accounting verified in the analysis. Codebook $1024 \times 1024 = 1{,}048{,}576$; tangent bases $1024 \times 1024 \times 16 = 16{,}777{,}216$; total trainable about 45.34 M of 348.53 M (13.0%).

Adopted / not adopted. Nothing was implemented. The 8-frame training / 32-frame testing protocol of this paper is the origin of the "8 frames" setting mentioned in the project; this project trains and tests on 32 frames per video and subsamples 8 frames only for validation monitoring (Section 6.2). The codebook idea is the only codebook in the project record, and it was never implemented (Section 5.4).

3.4 Paper 6 — CA-SOADD (Zhao et al.)

Citation. Y. Zhao, K. Zhang, Z. Hua, Y. Zhang, Q. Liao, W. Jiang, "Learning Tight Rejection Boundaries without Negatives for Strict One-Class Audio Deepfake Detection", ICML 2026 (PMLR) [4].

Method. Strict one-class audio detection: real-only training, a non-learnable running centroid c, compactness $1 - \cos(z, c)$, benign-invariance, and a centroid-anchored margin $\max(0, m - \cos(z, c) + \cos(z^-, c))$ against structure-disrupting probes z^-; online standardisation; leave-one-family-out evaluation.

Adopted / not adopted. Not implemented. Its leave-one-family-out protocol motivated the leave-one-generator-out evaluation of M5. A real-only image variant (candidate C3) was proposed in the analysis but not pursued.

3.5 Papers 7/8 — LNCLIP-DF / GenD (Yermakov et al.)

Citations. A. Yermakov, J. Cech, J. Matas, M. Fritz, "Deepfake Detection that Generalizes Across Benchmarks", arXiv:2508.06248, 2025 [5] (the M1 notebook records the venue as WACV 2026 and the code as `github.com/yermandy/GenD`); and the earlier version A. Yermakov, J. Cech, J. Matas, "Unlocking the Hidden Potential of CLIP in Generalizable Deepfake Detection", arXiv:2503.19683 [6].

Method. Frozen CLIP ViT-L/14; only LayerNorm affine parameters and a linear head on the L2-normalised CLS token are trained (104,450 parameters, 0.03%); loss $L_{\text{CE}} + 0.1L_{\text{align}} + 0.5L_{\text{unif}}$ (alignment/uniformity of Wang & Isola [12]); optional slerp latent augmentation; paired real/fake training data; RetinaFace crops with landmark alignment at 256 px.

Reported results used as reference. arXiv v1 Table 3 (video AUROC): FF 98.0, DF 99.6, F2F 98.2, FS 98.6, NT 95.7, FaceShifter 88.2 (87.3 without slerp, Table 5). GenD CLIP Table 4: DFD 96.4, UADFV 99.2, FaceShifter 86.6 (means of 5 seeds).

Adopted. The whole baseline recipe (backbone, trainable set, loss, optimiser schedule, augmentations, real/fake pairing) is adopted unchanged in M1 and kept in every later experiment. **Not adopted.** RetinaFace alignment (MTCNN boxes used instead), the external validation mix (FF++ val used), slerp augmentation, multiple seeds.

3.6 Other works used by the method or found in novelty checks

- **Used directly:** FF++ [8]; FaceShifter [9]; DFD (Google/Jigsaw, distributed with FF++); UADFV [10]; MTCNN [11]; CLIP [7]; alignment/uniformity [12]; Fisher-weighted merging [21]; Kish effective sample size [22]; leave-one-domain-out protocol in the spirit of DomainBed [24].
- **Used as concepts:** intrinsic dimension / random-subspace training [14, 15]; seeded generation of weights (PRANC, NOLA) [16, 17]; shared-plus-specific decomposition [18, 19, 20]; random-effects meta-analysis [23] (an analogy, later dropped).
- **Found in novelty checks** (recorded in the project conversation; titles and IDs as recorded, content not re-verified for this report): S3 "Suppressing Forgery-Specific Shortcuts" (arXiv:2606.01843), FMSD (arXiv:2601.01041), RINE (arXiv:2402.19091), CSD, CDA (ICCV 2025), DF40 (NeurIPS 2024), "Not All Attacks Are Learned Equally in Speech Deepfake Detection" (arXiv:2609.11763), FedNova, "Diversity over Uniformity" (CVPR 2026), SAFE (AAAI). They shaped which claims were *not* made (e.g. gradient-subspace projection was already published in S3).

3.7 Comparison

Table 3: Literature comparison. "Adopted" = used unchanged; "inspired" = idea taken, implementation ours; "none" = not used.

Paper	Main approach	Backbone	Trainable	Key mathematical idea	What we took	Limitations (as analysed)
Tian et al. 2023	Unsupervised intra-/cross-modal inconsistency	identity, video and audio networks	all networks (count not analysed)	consistency of identity over time and of audio vs video	none	needs audio; motion-transfer fakes; FakeAVCeleb only
Mapping Networks (CVPR'26)	weights generated from a latent	small CNNs, ResNet50	latent z ($\sim 10^3$)	$\hat{\theta} = \sigma(Wz + b)$	inspired M2/M2b latent (failed)	stores $O(Pd)$ mapping; no cross-generator test
Breaking Manifold Continuity (ICML'26)	VQ codebook on real CLIP patch tokens + tangent residuals	CLIP ViT-L/14 + Effort	45.34 M (13.0%)	discrete bottleneck, $\|(I - U_k U_k^\top)\Delta\|$	none (8/32-frame protocol noted)	large trainable set; codebook size chosen on test sets
CA-SOADD (ICML'26)	strict one-class audio, centroid margin	frozen SSL speech encoder	not reported	$\max(0, m - \cos(z, c) + \cos(z^-, c))$	inspired LOGO protocol	audio only
LNCLIP-DF / GenD (arXiv'25, WACV'26)	LayerNorm tuning of frozen CLIP	CLIP ViT-L/14	104,450 (0.03%)	L2-normalised CLS; CE + align + uniform	**adopted** as baseline and backbone of GBLA	external val set overlaps test datasets; mechanism of slerp unverified

4 Evolution of the Research Architecture

4.1 Initial ideas (literature phase)

Six hypotheses (A–F) and five candidates (C1–C5) were written down on 24 September: an adapter-geometry study (LN vs LoRA at matched size), a training-free real-manifold residual, a real-only centroid detector, a latent-subspace adapter over LN space, and explicit generator-invariance (rejected as already published). None of C1–C3 or C5 was implemented. C4 (latent over LN space) became the first GBLA design and was implemented in M2/M2b.

4.2 Component decisions

Table 4: Every architectural component considered, with purpose, dimensions and implementation status.

Component	Purpose and reason	Input → output	Source	Status
Face crop (MTCNN, 1.3× square)	remove background; same box for real and fake	frame → $224 \times 224 \times 3$	[11]	**completed** (M0)
CLIP ViT-L/14 visual encoder (frozen, QuickGELU)	transferable features without fine-tuning cost	$3\times224\times224 \to \mathbb{R}^{1024}$ (CLS after `ln_post`)	[7]	**completed** (M1–)
CLIP text projection (1024×768)	not needed for a binary head	–	–	removed (786,432 params not counted)
LayerNorm affine (50 modules)	the only adapted backbone parameters; per-channel scale/shift cannot create new feature directions	$\gamma, \beta \in \mathbb{R}^{1024}$ each	[5]	**completed** (trainable)
L2 normalisation	hyperspherical features for align/uniform losses	$\mathbb{R}^{1024} \to S^{1023}$	[5]	**completed**
Linear head	real/fake logits	$1024 \to 2$ (2,050 params)	[5]	**completed**
Seeded latent $\theta = \theta_0 + \alpha W z$	compress trainable set to $d+1$ numbers	$z \in \mathbb{R}^d \to \theta \in \mathbb{R}^P$	[14, 16, 2]	**failed** (M2, M2b); removed
Shared θ_s + private δ_g (GBLA)	stop dominant generators from fixing the shared solution	$\theta_s, \delta_g \in \mathbb{R}^{104,450}$	our use of [19, 20]	**completed** (M4–)
Pure-generator round-robin batching	each step sees one generator and its own reals	block of 225 steps	ours	**completed** (M3 sampler, M4–)
LoRA adapters	–	–	–	**not implemented**
Codebook / vector quantisation	–	–	[3] (literature only)	**not implemented**
Slerp latent augmentation	–	–	[6]	not used

4.3 Why each major decision was taken

- **CLIP ViT-L/14 with LayerNorm tuning.** It was the strongest published parameter-efficient detector with exact, verifiable parameter accounting (104,450), and LayerNorm tuning cannot synthesise new channel directions, which was argued to limit memorisation of generator artifacts.
- **Latent re-parameterisation (tried, removed).** It was meant to reduce the deployed adaptation to $d+1$ numbers and to make generator balancing exact in a small space. M2/M2b showed that $d \leq 1024$ loses most of the adaptation (Section 9.3).
- **Shared plus private offsets.** M3 showed one generator (Face2Face) dominates pooled training and one (NeuralTextures) is almost ignored. Private offsets let each generator fit its own residual during training, while the deployed θ_s is shared.
- **No LoRA.** LNCLIP-DF reports that LoRA overfits quickly on this task; the project's hypothesis was that the structure of the adaptation (diagonal vs low-rank mixing) matters. LoRA was listed as a comparison (candidate C1) but never implemented.

5 Mathematical Foundations

Notation: x is a face crop, $y \in \{0, 1\}$ its label (1 = fake), $g \in \mathcal{G}$ the generator of a fake, $G = |\mathcal{G}|$. Bold or capital letters denote matrices; $P = 104{,}450$ is the number of adapted parameters.

5.1 A. Feature extraction with the frozen CLIP ViT-L/14

Pre-processing. A 224×224 RGB crop is scaled to $[0, 1]$ and normalised per channel with CLIP's mean $(0.48145466, 0.4578275, 0.40821073)$ and standard deviation $(0.26862954, 0.26130258, 0.27577711)$: $\tilde{x} = (x - \mu)/\sigma \in \mathbb{R}^{3\times224\times224}$.

Patch embedding. A convolution with kernel and stride 14 splits the image into $16 \times 16 = 256$ patches and maps each to 1024 dimensions:

$$E = \mathrm{Conv}_{14\times14}(\tilde{x}) \in \mathbb{R}^{1024\times16\times16} \;\Rightarrow\; [e_1, \ldots, e_{256}],\; e_i \in \mathbb{R}^{1024}, \tag{1}$$

with $1024 \cdot 3 \cdot 14 \cdot 14 = 602{,}112$ weights (no bias), matching the M1 parameter report.

Tokens and positions. A learned class token c is prepended and learned positional embeddings p_i are added, giving 257 tokens:

$$H^{(0)} = \mathrm{LN}_{\mathrm{pre}}\big([c; e_1; \ldots; e_{256}] + [p_0; \ldots; p_{256}]\big) \in \mathbb{R}^{257\times1024}, \tag{2}$$

with $1024 + 257 \cdot 1024 = 264{,}192$ parameters (class + positional), as reported.

Transformer blocks. For $l = 1, \ldots, 24$ (pre-norm):

$$U^{(l)} = H^{(l-1)} + \mathrm{MHSA}\big(\mathrm{LN}_1^{(l)}(H^{(l-1)})\big), \tag{3}$$

$$H^{(l)} = U^{(l)} + \mathrm{MLP}\big(\mathrm{LN}_2^{(l)}(U^{(l)})\big). \tag{4}$$

Multi-head self-attention with $h = 16$ heads of size $d_h = 64$:

$$\mathrm{head}_j = \mathrm{softmax}\Big(\frac{Q_j K_j^\top}{\sqrt{d_h}}\Big)V_j, \qquad Q_j = ZW_j^Q,\; K_j = ZW_j^K,\; V_j = ZW_j^V, \tag{5}$$

$$\mathrm{MHSA}(Z) = [\mathrm{head}_1; \ldots; \mathrm{head}_{16}]\, W^O. \tag{6}$$

In `open_clip` the three projections are one fused 3072×1024 input projection; per block attention has $3072 \cdot 1024 + 3072 + 1024 \cdot 1024 + 1024 = 4{,}198{,}400$ parameters ($\times 24 = 100{,}761{,}600$). The MLP is $1024 \to 4096 \to 1024$ with QuickGELU, $\mathrm{QuickGELU}(u) = u\,\sigma(1.702u)$, and 8,393,728 parameters per block ($\times 24 = 201{,}449{,}472$). These products reproduce the M1 report exactly. The number of heads (16) is the standard ViT-L/14 configuration; it is not printed in a notebook.

LayerNorm on a token $u \in \mathbb{R}^{1024}$:

$$\mathrm{LN}(u) = \gamma \odot \frac{u - \mathrm{mean}(u)}{\sqrt{\mathrm{var}(u) + \epsilon}} + \beta, \qquad \gamma, \beta \in \mathbb{R}^{1024}. \tag{7}$$

There are $2 \times 24 + 2 = 50$ LayerNorms (two per block, `ln_pre`, `ln_post`), so $50 \times 2048 = 102{,}400$ affine parameters. γ rescales and β shifts each channel; they cannot mix channels.

Feature. The representation is the class token after the final LayerNorm; the CLIP projection to the text space is removed:

$$f(x) = \mathrm{LN}_{\mathrm{post}}\big(H_0^{(24)}\big) \in \mathbb{R}^{1024}, \qquad z(x) = \frac{f(x)}{\|f(x)\|_2} \in S^{1023}. \tag{8}$$

5.2 B. Parameter-efficient fine-tuning

Why full fine-tuning is expensive. Training all 303,181,826 parameters requires gradients and two Adam moments for each (about 3× the weights in extra memory) and gives the model capacity to memorise generator artifacts.

LayerNorm tuning (implemented). The trainable set is $\theta = \{\gamma^{(m)}, \beta^{(m)}\}_{m=1}^{50} \cup \{W_h, b_h\}$ with $|\theta| = 102{,}400 + 2{,}050 = 104{,}450$; everything else is frozen. The update of a LayerNorm is a diagonal affine map in each layer.

LoRA (not implemented; explained because it was requested). LoRA [13] keeps a frozen weight $W \in \mathbb{R}^{d_{\mathrm{out}}\times d_{\mathrm{in}}}$ and learns a low-rank update

$$W' = W + \Delta W = W + \tfrac{\alpha}{r} BA, \qquad B \in \mathbb{R}^{d_{\mathrm{out}}\times r},\; A \in \mathbb{R}^{r\times d_{\mathrm{in}}}, \tag{9}$$

with $r(d_{\mathrm{in}} + d_{\mathrm{out}})$ trainable parameters per adapted matrix. For rank r on separate query and value projections (1024×1024) of all 24 blocks,

$$N_{\mathrm{LoRA}} = 24 \times 2 \times r\,(1024 + 1024) = 98{,}304\, r. \tag{10}$$

For $r = 8$ this gives 786,432 parameters in $24 \times 2 \times 2 = 96$ tensors. This arithmetic matches the supplied reference values (786,432 trainable, 96 tensors, `q_proj/v_proj`), but no LoRA code, configuration or output exists in this project. `q_proj/v_proj` are module names of the Hugging Face CLIP implementation; this project uses `open_clip`, whose attention has a fused projection. See Section 9.10.

5.3 C. Latent re-parameterisation (implemented in M2/M2b, failed)

Motivation: optimise in $\mathbb{R}^d$, $d \ll P$, so that the deployed adaptation is $d + 1$ numbers.

$$\theta(z, \alpha) = \theta_0 + \alpha\, W z, \qquad W \in \mathbb{R}^{P \times d},\ W_{ij} \sim \mathcal{N}(0, 1/d) \text{ (regenerated from a seed, never stored)}, \tag{11}$$

with $z \in \mathbb{R}^d$ initialised at 0 and α at 1. The gradient is $\nabla_z L = \alpha W^\top \nabla_\theta L$ (verified numerically in M2/M2b to a relative difference of 3.3×10^{-7} and 1.1×10^{-6}). M2 applied (11) to LN + head ($P = 104{,}450$); M2b only to the LayerNorms ($P = 102{,}400$) with a directly trained head. This is a linear random-subspace method [14]; unlike Mapping Networks [2] it has no non-linearity and no Mapping Loss. It is *not* part of the current architecture.

5.4 D. Codebook and quantisation

No codebook, vector quantisation or codebook-utilisation measurement exists in this project's implementation. The only codebook in the project record is that of Breaking Manifold Continuity [3], where patch tokens z_i are assigned to the nearest code, $k(i) = \arg\min_k \|z_i - e_k\|$, and trained with a VQ commitment loss. Utilisation (the fraction of the K codes that receive at least one assignment) was never measured here. The reference values 97/1024 (9.47%) and 46/1024 (4.49%) are therefore **unverified** (Section 9.10).

5.5 E. Classification and loss

Head and prediction.

$$\ell(x) = W_h z(x) + b_h \in \mathbb{R}^2, \quad W_h \in \mathbb{R}^{2 \times 1024}, \qquad p(x) = \mathrm{softmax}(\ell(x))_1 \text{ (fake probability)}. \tag{12}$$

Loss (as in LNCLIP-DF and its code), for a batch $\mathcal{B}$:

$$L = L_{\mathrm{CE}} + 0.1\, L_{\mathrm{align}} + 0.5\, L_{\mathrm{unif}}, \tag{13}$$

$$L_{\mathrm{CE}} = -\tfrac{1}{|\mathcal{B}|} \textstyle\sum_i \log \mathrm{softmax}(\ell(x_i))_{y_i}, \tag{14}$$

$$L_{\mathrm{align}} = \mathbb{E}_{(i,j):\, y_i = y_j,\, i \neq j} \|z_i - z_j\|_2^2, \qquad L_{\mathrm{unif}} = \log\, \mathbb{E}_{i \neq j}\, e^{-2\|z_i - z_j\|_2^2}. \tag{15}$$

Alignment pulls same-class features together; uniformity spreads all features over the sphere, which prevents collapse to two points [12]. The total loss is negative in practice because $L_{\mathrm{unif}} < 0$ (e.g. M1 epoch 20: loss -1.64, CE 0.143, align 1.973, uniform -3.966).

Learning rate (function of the global step t, steps per epoch $S = 900$):

$$\eta(t) = \begin{cases} \eta_{\min} + (\eta_{\max} - \eta_{\min})\, \dfrac{t'}{S}, & t' < S, \\ \eta_{\min} + \tfrac{1}{2}(\eta_{\max} - \eta_{\min})\big(1 + \cos(\pi\, \tfrac{t' - S}{10S - S})\big), & \text{otherwise}, \end{cases} \qquad t' = t \bmod 10S, \tag{16}$$

with $\eta_{\max} = 3 \times 10^{-4}$, $\eta_{\min} = 10^{-5}$: one warm-up epoch, then cosine decay inside 10-epoch cycles.

5.6 F. GBLA: shared and private parameters

During training each generator g uses its own parameters, and only the shared part is kept:

$$\theta_g = \theta_s + \delta_g, \qquad \min_{\theta_s, \{\delta_g\}} \sum_{g \in \mathcal{G}} \Big[L\big(\theta_s + \delta_g; \mathcal{D}_g\big) + \lambda \|\delta_g\|_2^2 \Big], \tag{17}$$

where $\mathcal{D}_g$ contains all training fakes of g and a fixed, disjoint $1/G$ share of the training reals (round-robin assignment). Each step uses a pure batch from one g. With $G_t = \nabla_\theta L(\theta_s + \delta_g; \mathcal{B}_t)$:

$$\theta_s \leftarrow \mathrm{Adam}(\theta_s,\, G_t), \qquad \delta_g \leftarrow \mathrm{Adam}(\delta_g,\, G_t + 2\lambda\delta_g), \tag{18}$$

with separate Adam moments for θ_s and each δ_g. $\lambda = \infty$ keeps $\delta_g = 0$ (pooled training with the same batching); $\lambda = 0$ gives G unregularised offsets. At test time $p(x)$ is computed with θ_s only. Training stores $(G+1)P = 522{,}250$ numbers for $G = 4$; $P = 104{,}450$ are deployed.

Intuition. In pooled training a generator whose artifacts are easy to fit moves the shared parameters most. A private offset absorbs the part of each generator's optimum that the others do not share, so the shared solution is less determined by the dominant generator. λ sets how much each generator may deviate.

5.7 G. Generator dominance: quadratic model and G_{eff}

Hypothesis tested in M3 (failed). Near each generator's own optimum v_g, $L_g(\theta) \approx \frac{1}{2}(\theta - v_g)^\top H_g(\theta - v_g)$. Pooled training then gives a curvature-weighted average, and the GBLA objective under the same approximation gives

$$\theta^* \approx \Big(\sum_g H_g\Big)^{-1} \sum_g H_g v_g, \qquad \theta_s \approx \Big(\sum_g A_g\Big)^{-1} \sum_g A_g v_g, \quad A_g = (H_g + 2\lambda I)^{-1} H_g. \tag{19}$$

As $\lambda \to \infty$, $A_g \propto H_g$ (pooled); as $\lambda \to 0$, $A_g \to I$ (equal weights). M3 measured 4×4 Hessians in the span of the four solutions; they were asymmetric ($\|H - H^\top\|/\|H\|$ up to 0.55), step-size dependent and indefinite, so this model was dropped and the random-effects reading $\tau^2 = 1/(2\lambda)$ is not used.

Geometric dominance (kept). With U an orthonormal basis of $\{v_g - \theta_0\}$, solve $\theta^* - \theta_0 \approx \sum_g w_g(v_g - \theta_0)$ in that span and normalise the weights to sum to 1. Kish's effective number [22]

$$G_{\text{eff}} = \frac{(\sum_g w_g)^2}{\sum_g w_g^2} \in [1, G] \tag{20}$$

is G for equal weights and 1 when one generator carries all weight.

5.8 H. Merge baselines

Uniform: $\theta = \frac{1}{|S|}\sum_{g \in S} v_g$. Diagonal Fisher [21]: $\theta_i = \sum_g F_{g,i} v_{g,i} / (\sum_g F_{g,i} + \epsilon)$, $F_g = \mathbb{E}[(\nabla_\theta L_g)^2]$ estimated on 30 fixed batches of 100 fakes of g and 25 reals.

5.9 I. Evaluation

Frame AUC uses every crop. Video score is the mean fake probability over the video's frames (32 for FF++, DFD, UADFV), $s(v) = \frac{1}{|v|}\sum_{x \in v} p(x)$; video AUC is the Mann–Whitney AUC of these scores. For each FF++ generator its 140 test fake videos are compared with the same 140 real test videos. Accuracy uses threshold 0.5. Cross-dataset confidence intervals use a cluster bootstrap (2,000 resamples) over actors (DFD, 28 clusters) or real/fake pairs (UADFV, 49 clusters); paired differences use the same resamples for both models.

5.10 J. The complete implemented pipeline

$$\boxed{p(\text{fake} \mid v) = \frac{1}{32}\sum_{t=1}^{32} \text{softmax}\Big(W_h \frac{f_{\theta_s}(\mathcal{N}(\text{crop}(F_t)))}{\|f_{\theta_s}(\cdot)\|_2} + b_h\Big)_1} \tag{21}$$

F_t: frame t of video v; crop: MTCNN box, 1.3× square, 224×224; $\mathcal{N}$: CLIP normalisation; f_{θ_s}: frozen ViT-L/14 whose 50 LayerNorms take the values in θ_s; $W_h, b_h \in \theta_s$. Training adds δ_g and the penalty of (17); inference uses θ_s only. No LoRA, codebook or latent mapping is present.

6 Datasets and Preprocessing

6.1 FaceForensics++ c23

The data on C-DAC is a Kaggle copy of FF++ at compression c23, stored under `ffc23dataset/kaggledata/FaceForensics++_C23`. It has the folders `csv`, `DeepFakeDetection`, `Deepfakes`, `Face2Face`, `FaceShifter`, `FaceSwap`, `NeuralTextures` and `original`, and a sibling folder `frames_all` with 32 pre-extracted frames per video.

Table 5: FF++ c23 categories used (M0 section 2.1). Example resolutions are from the first video of each category; per-dataset resolution and FPS were not tabulated for FF++.

Category	Videos	Frame folders	Frames/video	Notes
original	1000	1000	32	real; first video 640×480
Deepfakes (DF)	1000	1000	32	face swap (autoencoder); named `target_source`
Face2Face (F2F)	1000	1000	32	expression re-enactment
FaceSwap (FS)	1000	1000	32	graphics-based face swap
NeuralTextures (NT)	1000	1000	32	neural rendering of the mouth region
FaceShifter (FSh)	1000	1000	32	face swap; **never trained** except in LOGO folds and M6/M7
DeepFakeDetection	1000 entries	–	–	not used from this copy; full DFD obtained separately (M8a)

6.2 Frames

- **FF++:** the 32 frames per video in `frames_all` were used (`FRAME_SOURCE="frames"`; all 1,002 box files record source "frames"). File names are `<video>_frame_NN.jpg`; the frame key is $NN \in \{0, \ldots, 31\}$. M8a measured their encoding: JPEG quality 95 with 4:2:0 chroma subsampling.
- **DFD and UADFV:** 32 evenly spaced frames decoded from each video, followed by a JPEG round trip at quality 95, 4:2:0, so that test frames carry the same compression as the training frames (M8a checked that this lowers the pixel error against `frames_all`).
- **8 vs 32 frames.** Training and testing always use all 32 frames per video. From M2 onward the *validation* set keeps 8 evenly spaced frames per video (5,600 instead of 22,400 crops) only to monitor training; no model or reported test number depends on it. The 8-frame training protocol of Breaking Manifold Continuity was not used.

6.3 Face detection and cropping

- Detector: MTCNN from `facenet-pytorch` 2.6.0 (`keep_all=True`, `post_process=False`), largest face with probability ≥ 0.9; frames without a confident face take the box of the nearest frame that has one.
- Crop: square of side $1.3 \times \max(w, h)$ (clipped to the frame), resized to 224×224 (bicubic), saved as lossless PNG.
- FF++ fakes are cropped with the *original video's box for the same frame key*, so real and fake crops cover the same region ("frames with nearest box": 0). DFD and UADFV fakes are detected independently because they are not frame-aligned with their real videos.
- **RetinaFace.** LNCLIP-DF uses RetinaFace with landmark alignment at 256 px. This project used MTCNN without alignment at 224 px from the start; no RetinaFace installation attempt and no manual OpenCV bounding-box experiment is recorded in any file, so none is reported.



Figure 1: M0 smoke test: frame 16 of original video 000 and of its five manipulated versions (`000_003`), all cropped with the original's box. The same region appears in every panel.

6.4 Splits and index

The official FF++ split lists (360/70/70 video pairs) give 720/140/140 original videos per split; a pair $[a, b]$ puts originals a, b and fakes a_b, b_a in the same split. The index `index/ffpp_c23.csv` has 192,000 rows (one per crop) with columns `split, label, generator, video_id, target_id, source_id, frame_idx, path, pair_path`; 100% of fake crops have a paired real crop.

Table 6: Summary Table 6 — dataset statistics (crops; videos in brackets).

Set	Train	Val (monitoring)	Test	Use
FF++ original (real)	23,040 (720)	4,480 (140); 1,120 from M2	4,480 (140)	all experiments
FF++ each of DF, F2F, FS, NT	23,040 (720)	4,480 (140); 1,120 from M2	4,480 (140)	training generators
FF++ FaceShifter	0 (23,040 in M5–M7 folds)	0	4,480 (140)	unseen generator
FF++ training set (M1)	**115,200** (real:fake 1:4)	22,400	26,880	
DFD (M8a)	–	–	11,616 real (363) + 97,838 fake (3,068)	cross-dataset
UADFV (M9a)	–	–	1,568 real (49) + 1,568 fake (49)	cross-dataset

6.5 Cross-dataset sets

DFD (M8a). 363 real and 3,068 fake videos, 28 actors, 16 scenes, all 1920×1080 at 24 fps; real duration mean 36.2 s, fake 30.5 s; 19 videos have fewer than 32 frames (crops per video: min 5, median 32). 109,454 crops; 504 frames without a face; 19.4% of frames have two or more confident faces (three scenes almost always two-person).

UADFV (M9a). 49 real and 49 fake videos (each fake made from the real video with the same number), 30 fps, resolutions from 234×268 to 536×500, mean duration 11.8 s. 3,136 crops (32 per video); 21 frames without a face; 99 frames (3.2%) with ≥ 2 faces. Each folder also contained `frames/` and `landmarks/` directories, which explains the 51 entries counted before M9a.

7 Implementation Details

7.1 Environment and organisation

Python 3.10.12, PyTorch 2.6.0+cu124 (CUDA 12.4), torchvision 0.21.0+cu124, open_clip 3.3.0, OpenCV 5.0.0, NumPy 2.2.6, pandas 2.3.3, facenet-pytorch 2.6.0, Pillow 12.3.0, on an NVIDIA A100-SXM4-40GB (C-DAC on-demand Jupyter). One notebook per milestone; each notebook reads the M0 index and crops, writes runs to `runs/<milestone>/` on C-DAC, and only the notebook (with its outputs) is committed. Model checkpoints never enter git.

Table 7: Summary Table 1 — model configuration (all trained models).

Item	Value	Evidence
Backbone	OpenAI CLIP ViT-L/14 visual tower, `open_clip`, `force_quick_gelu=True`	M1 code
Hidden width / MLP width	1024 / 4096	M0 table; M1 counts
Transformer blocks	24	50 LNs $= 2 \cdot 24 + 2$; counts
Attention heads	16 (standard config)	not printed
Patch size / input / tokens	14 / 224×224 / 257	patch and position counts
Representation	CLS after `ln_post`, L2-normalised, 1024-d; projection removed	M1 3.1
Classifier	Linear($1024 \to 2$), 2,050 params	M1 3.2
Trainable	50 LayerNorms (102,400) + head (2,050) = 104,450	M1 3.2
LoRA	none	–
Loss	CE + 0.1 align + 0.5 uniform	M1 code
Optimiser	AdamW, $\beta = (0.9, 0.999)$, weight decay 0	M1 code
Learning rate	warm-up 1 epoch $10^{-5} \to 3 \times 10^{-4}$, cosine to 10^{-5}, 10-epoch cycles	M1 code
Batch / precision	128 / bf16 autocast	M1 code
Epochs	M1: 20 (2 cycles); all later runs: 10	notebooks
Steps per epoch	900	M1 2.2, M4 2.3
Augmentation	h-flip 0.5; affine (10°, translate 0.1, scale 0.9–1.1); Gaussian blur k=7, $\sigma \in [0.1, 2]$, p=0.1; colour jitter (0.1, 0.1); JPEG quality 40–100	M1 code
Seed	0 (single seed everywhere)	configs
GBLA	$\lambda \in \{\infty, 1, 0.1, 0.01, 0.001, 0\}$; chosen $\lambda = 0.01$	M4

7.2 Data flow and checkpoints

- **M0** writes `faces/<category>/<video>/<key>.png`, `faces/_boxes/<id>.json` and the index CSV. Extraction is resume-safe (per-video `_done`/`_failed` markers, boxes saved before crops, atomic writes, error log).
- **M1**–**M7** load crops through the index; from M2 on, all crops are decoded once into RAM (train 17.3 GB; 20.8 GB with FaceShifter) to remove file-system I/O.
- **Checkpoints** contain only the trainable numbers (and optimiser state in `last.pt`); one per epoch (`epoch_XX.pt`), written atomically; runs resume after the last finished epoch with the same seeded data order. Settings are stored in `config.json`; a run refuses to resume if they changed.
- **Model selection.** No checkpoint is selected on a test set. M1 reports the final epoch (20) and, for reference, the best FF++-validation epoch (7). All later models use the final epoch (10).
- **M8b/M9b** load the nine finished checkpoints, first reproduce saved FF++ predictions (max $|\Delta| = 0.0$ for all GBLA runs; FaceShifter AUC of merges and M1 within tolerance), then write per-crop predictions `runs/m8b|m9b/pred_<model>.csv`.

Table 8: Summary Table 7 — hardware and runtime (as printed in the notebooks).

Item	Value
GPU	NVIDIA A100-SXM4-40GB (42.3 GB reported)
CPUs usable by the job	1–16 depending on the session (node has 256)
M1 training speed / peak memory	253 img/s; 28.9 GB; no gradient checkpointing
M1 training time	17.3–19.5 min per epoch; 363 min (6.05 h) for 20 epochs (sum of logged epoch times)
Latent / GBLA runs (estimates printed)	M2: 7.6 min/epoch, 1.3 h per d; M4: 1.3 h per λ; M3 per-generator: 1.9 min/epoch
M6 FaceShifter-only run	2.0 min per epoch (logged), 10 epochs
Crops into RAM	0.9–5.7 min for 115,200 crops (2,206 crops/s with 16 CPUs in M3; 339 crops/s with 1 CPU in M2b)
M8a DFD extraction	127 min for 3,431 videos
M9a UADFV extraction	1.8 min for 98 videos
Prediction	DFD: about 169 s per model (109,454 crops); UADFV: 5 s per model

8 Experimental Setup

- **Standard protocol.** Train on FF++ c23 train split, real + DF, F2F, FS, NT; test on the FF++ test split with all five generators; FaceShifter is the unseen generator.
- **Leave-one-generator-out (M5).** Train on four of the five FF++ generators, test on the fifth; all five folds.
- **Subsets (M7).** All 25 subsets of size 2, 3, 4 of the five generators.
- **Cross-dataset (M8b, M9b).** Models trained under the standard protocol, tested without any retraining on DFD and UADFV.
- **Budget.** 10 epochs × 900 steps × 128 images for every model compared in M4–M9b. M1 has 20 epochs; its epoch-10 checkpoint is used as the matched-budget reference.
- **Metric.** Video AUC (primary), frame AUC, accuracy at 0.5, AP and EER for cross-dataset.

9 Results and Analysis

9.1 M0 — data pipeline and parameter check (completed)

All 6,000 videos extracted with 0 failures, 32 crops per video, 192,000 crops, 100% pairing. The parameter report gave the ViT-L/14 tower at 303,966,208 parameters *including* the 786,432-parameter final projection, and the adapted set at 104,450 (Table 19).

9.2 M1 — LNCLIP-DF reproduction (completed)

Integrity check before training. Paired fake/real pixel difference (median) 4.80–15.88 vs 54.63–57.43 for a random real; the paired crop is closer for 99.01% (F2F) to 100% of 600 sampled fakes: `PAIRING OK`.

Table 9: Summary Table 3 — M1 epoch-wise training record (FF++ val = real + four training generators, 32 frames). Accuracy and validation loss were not logged per epoch.

Ep	LR (end)	Loss	CE	Align	Unif	CE real	CE NT	Val frame AUC	Val video AUC	min
1	3.0e-4	−0.853	0.568	1.603	−3.162	0.916	0.518	83.08	85.44	18.5
2	3.0e-4	−1.357	0.379	1.928	−3.857	0.846	0.473	92.83	95.87	18.5
3	3.0e-4	−1.468	0.291	1.943	−3.906	0.663	0.418	94.69	97.23	18.5
4	2.0e-4	−1.515	0.252	1.950	−3.925	0.589	0.378	95.56	97.83	18.6
5	2.0e-4	−1.541	0.231	1.956	−3.935	0.551	0.346	95.69	97.84	18.8
6	1.0e-4	−1.560	0.215	1.959	−3.942	0.517	0.328	95.90	98.00	19.5
7	1.0e-4	−1.570	0.207	1.961	−3.946	0.503	0.320	96.03	**98.07**	19.1
8	0.0	−1.578	0.200	1.962	−3.949	0.486	0.309	95.93	97.97	18.2
9	0.0	−1.582	0.197	1.963	−3.950	0.480	0.305	95.94	97.95	18.1
10	0.0	−1.583	0.196	1.963	−3.950	0.478	0.301	95.94	97.95	18.1
11	3.0e-4	−1.585	0.194	1.964	−3.951	0.479	0.296	96.03	98.01	18.0
12	3.0e-4	−1.595	0.186	1.967	−3.955	0.464	0.282	96.04	97.99	18.0
13	3.0e-4	−1.608	0.175	1.969	−3.958	0.436	0.273	95.95	97.86	17.9
14	2.0e-4	−1.618	0.166	1.971	−3.961	0.420	0.256	96.11	97.98	17.8
15	2.0e-4	−1.628	0.156	1.971	−3.963	0.397	0.242	95.88	97.91	17.7
16	1.0e-4	−1.631	0.154	1.972	−3.964	0.395	0.239	96.05	97.90	17.4
17	1.0e-4	−1.637	0.149	1.973	−3.965	0.381	0.230	96.09	97.98	18.0
18	0.0	−1.640	0.145	1.973	−3.965	0.370	0.229	96.09	97.95	17.7
19	0.0	−1.640	0.146	1.973	−3.966	0.373	0.228	96.07	97.93	17.4
20	0.0	−1.642	0.143	1.973	−3.966	0.368	0.221	96.07	97.94	17.3

LR column as printed (rounded to one significant digit, so "0.0" means $\leq 5 \times 10^{-5}$). Final per-generator training CE: real 0.368, DF 0.017, F2F 0.071, FS 0.040, NT 0.221 (about 13× spread between DF and NT).

Table 10: M1 test results (video AUC %, frame AUC in brackets) and comparison with the paper.

Test set	Epoch 20 (final)	Epoch 10 (matched budget)	Epoch 7 (best val)	Paper (arXiv v1, Tab. 3)
Seen combined	98.04 (96.36)	97.60 (95.66)	97.67 (95.77)	98.0
Deepfakes	99.39 (98.40)	99.22 (98.17)	99.29	99.6
Face2Face	98.40 (96.78)	98.09 (96.06)	98.12	98.2
FaceSwap	99.38 (98.54)	99.20 (98.15)	99.20	98.6
NeuralTextures	94.98 (91.71)	93.88 (90.25)	94.06	95.7
FaceShifter (unseen)	**88.37** (81.98)	**88.19** (81.20)	88.37	88.2
Video acc. @0.5, FaceShifter	73.57	73.21	76.43	–

Observation. The reproduction is within −0.72 to +0.78 points of the paper on every test set. Validation AUC saturates after about epoch 6, so FF++ validation does not discriminate between models; this is a known property of FF++ and the reason it is used only for monitoring.

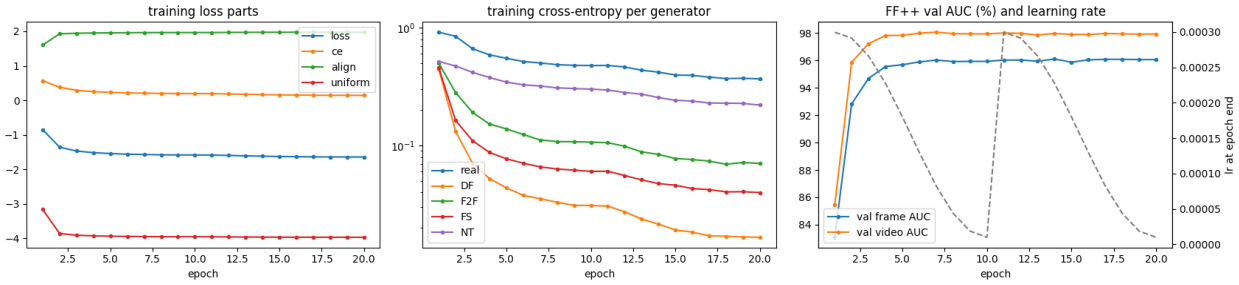


Figure 2: M1 training curves: loss parts (left), training cross-entropy per generator (middle), FF++ validation AUC and learning rate (right).

9.3 M2 and M2b — latent re-parameterisation (failed)

Table 11: Latent size sweep, 10 epochs each (video AUC %). M2: LN + head inside the latent; M2b: LN in the latent, head trained directly.

Model	Trainable	Seen	NT	FaceShifter	Note
M1 full LN tuning, epoch 10	104,450	97.60	93.88	88.19	reference
M2 latent $d = 16$	17	50.11	52.47	58.29	
M2 latent $d = 64$	65	46.56	49.40	55.56	learned the class ratio only
M2 latent $d = 256$	257	55.97	51.36	48.96	
M2 latent $d = 1024$	1,025	72.63	58.63	67.58	
M2 latent $d = 4096$	4,097	96.80	90.76	78.72	seen recovered, unseen not
M2b head only	2,050	83.71	58.96	77.65	reference
M2b latent LN $d = 64$	2,115	81.24	58.16	74.40	below head only
M2b latent LN $d = 256$	2,307	77.28	57.40	68.64	below head only
M2b latent LN $d = 1024$	3,075	80.81	60.73	75.16	below head only
M2b latent LN $d = 4096$	6,147	91.07	75.11	81.54	recovers 53% (seen), 37% (FSh) of the gap

Diagnosis. In M2, with $d \leq 256$ the training cross-entropy was about 1.2–1.5 on reals and 0.24–0.37 on every fake, i.e. the model predicted "fake" for nearly everything (the 1:4 class ratio). The head needs one specific direction in 1024-d space, but in the latent each direction moves all 104,450 parameters at once, about 2% of them head weights. The learning-rate probe in M2 selected 10^{-2} on total loss, which was dominated by the uniformity term. M2b moved the head out of the latent and probed on cross-entropy (chosen 3×10^{-4}, at the edge of the grid); no latent learning rate beat the head alone in the probe, and only $d = 4096$ beat the head alone after training.

Decision. The compact deployed payload (65 numbers + seed) was abandoned. The cost of the latent (W with up to 4.3×10^8 entries, regenerated every session) gives no benefit at the sizes where it would matter. From M3 on, GBLA operates directly on the 104,450 parameters.

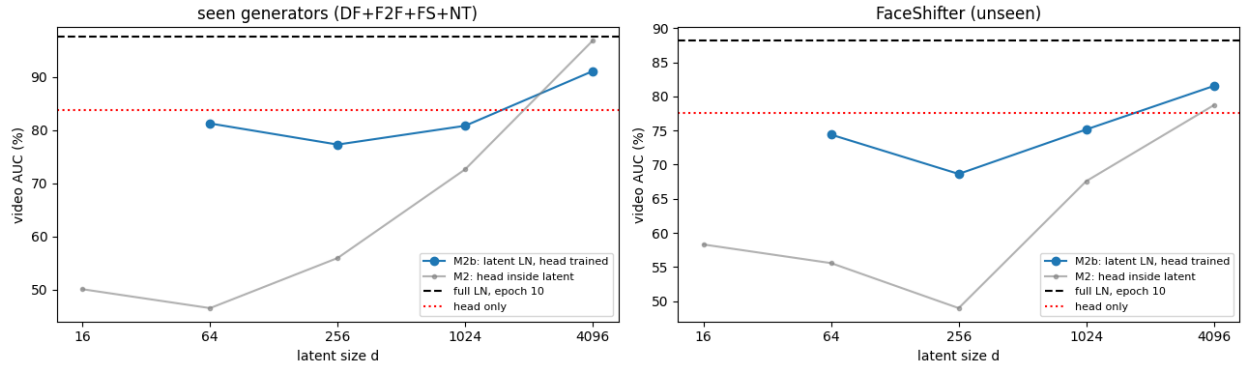


Figure 3: Video AUC against latent size d for M2 (grey) and M2b (blue), with full LN tuning (dashed) and the head alone (dotted). Left: seen generators; right: FaceShifter.

9.4 M3 — E1: generator dominance (curvature failed, geometry completed)

Each training generator was fitted alone (all of its fakes plus a fresh random quarter of the reals per epoch, 225 steps, 10 epochs).

Table 12: Cross-generator video AUC (%): rows = training data, columns = test generator.

Trained on	DF	F2F	FS	NT	FSh (unseen)
DF only	**99.91**	79.64	96.14	55.25	81.80
F2F only	98.67	**98.93**	99.20	64.88	83.80
FS only	99.03	86.07	**99.82**	51.07	**89.56**
NT only	87.68	69.15	80.62	**72.15**	80.65
Pooled (M1, epoch 10)	99.22	98.09	99.20	93.88	88.19

FaceSwap alone beats the pooled model on FaceShifter (89.56 vs 88.19). NeuralTextures is learnable only when pooled (72.15 alone vs 93.88 pooled).

Geometry. Distances $\|v_g - \theta_0\|$: DF 18.24, F2F 18.11, FS 18.28, NT 15.01, pooled 25.83. 64.5% of $\|\theta^* - \theta_0\|^2$ lies in the span of the four directions. Normalised weights: DF 0.184, **F2F 0.576**, FS 0.271, NT −0.031; $G_{\text{eff}} = 2.27$ of 4 ($G_{\text{eff}}/G = 0.569$).

Curvature model. Curvature-predicted weights (DF 0.103, F2F 0.891, FS 0.029, NT −0.023) did not match the geometric ones (max difference 0.315, different ranking); Hessians were asymmetric (0.13–0.55), changed with the finite-difference step (relative difference 0.36–0.62) and had negative eigenvalues; regrets were negative. Verdict as printed: **NO-GO** for the curvature explanation. The project continued on the geometric dominance result, not on the curvature theory.

9.5 M4 — GBLA λ sweep (completed)

Table 13: GBLA, deployed θ_s, 10 epochs (video AUC %) and final offset norms $\|\delta_g\|$.

λ	Seen	DF	F2F	FS	NT	FSh	$\|\delta\|$ DF / F2F / FS / NT
∞ (pooled, GBLA batching)	96.41	99.72	98.66	99.34	87.90	90.53	0 / 0 / 0 / 0
1	96.06	99.67	98.66	99.41	86.49	92.06	0.05 / 0.04 / 0.08 / 0.12
0.1	95.66	99.61	98.80	99.34	84.90	92.48	0.24 / 0.26 / 0.32 / 0.42
0.01	95.55	99.43	98.55	99.17	85.06	**92.91**	0.82 / 1.32 / 1.01 / 1.49
0.001	93.27	99.60	97.73	99.49	76.25	86.35	3.65 / 4.76 / 3.91 / 4.54
0	90.14	99.34	94.14	99.38	67.69	80.04	8.16 / 8.35 / 7.90 / 7.46
M1 pooled, epoch 10	97.60	99.22	98.09	99.20	93.88	88.19	–

Reading. Two effects must be separated. (i) Batching: pooled training with pure-generator blocks ($\lambda = \infty$) already scores 90.53 on FaceShifter vs 88.19 for M1's mixed batches, at the cost of NT (87.90 vs 93.88). (ii) Offsets: from $\lambda = \infty$ to $\lambda = 0.01$, FaceShifter gains +2.38, seen AUC loses 0.86 and NT 2.84. Too small λ (≤ 0.001) lets the offsets absorb the task and θ_s degrades.

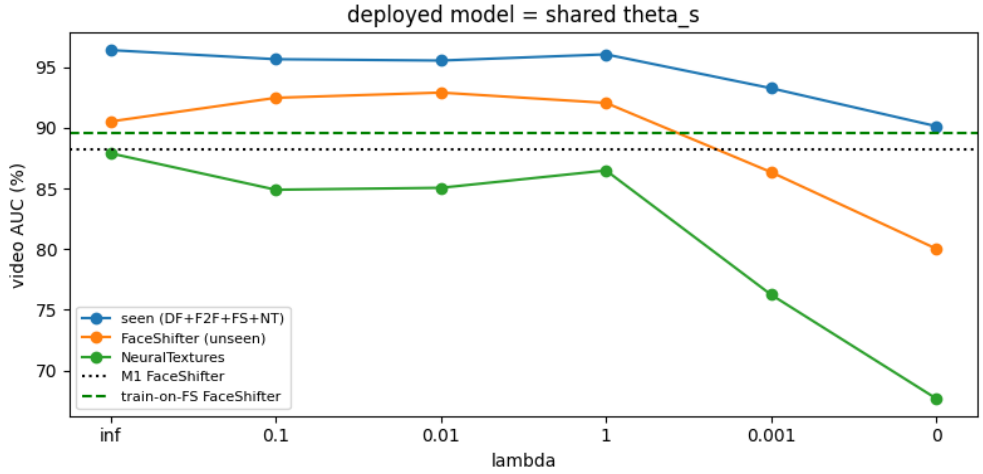


Figure 4: M4: seen, FaceShifter and NeuralTextures video AUC of θ_s against λ (x-axis in run order), with M1 FaceShifter (dotted) and the FaceSwap-only model on FaceShifter (dashed).

9.6 M5 and M6 — leave-one-generator-out and merge baselines (completed)

Table 14: Leave-one-generator-out, held-out video AUC (%) and seen AUC of the other four.

Held out	Pooled ($\lambda = \infty$)	GBLA ($\lambda = 0.01$)	Fisher merge	Uniform merge	Pooled seen	GBLA seen	Fisher seen
NT	**64.40**	62.29	57.09	53.34	99.15	99.27	97.21
FS	97.90	98.37	**98.71**	96.54	95.01	94.58	87.91
DF	99.21	99.22	**99.59**	98.96	94.42	94.61	87.39
F2F	85.40	**87.40**	82.33	75.03	91.94	92.77	89.21
FSh	90.53	**92.91**	88.99	81.43	96.41	95.55	89.24
Mean	87.49	**88.04**	85.34	81.06	95.39	95.36	90.19

GBLA − pooled = +0.55 (gain on FSh +2.38, F2F +2.00; loss on NT −2.11); GBLA − Fisher = +2.70; Fisher − pooled = −2.15. The FaceShifter-only model trained in M6 (10 epochs, 2.0 min/epoch, final CE 0.126) completes the five single-generator solutions.

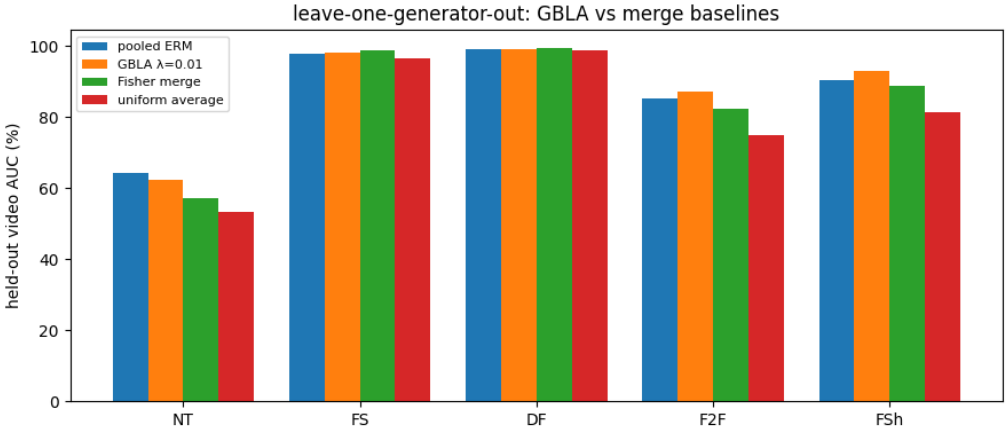


Figure 5: M6: held-out video AUC per fold for pooled ERM, GBLA ($\lambda = 0.01$), Fisher merge and uniform average.

9.7 M7 and M7b — the G_{eff} law (completed; law not supported)

Pooled models were trained on all 25 subsets (20 new runs; 5 reused from M4/M5). Raw Pearson correlation between $G_{\text{eff}}/|S|$ and mean OOD AUC: −0.768, the opposite of the hypothesis. M7b (no training) showed this is a composition effect: NeuralTextures is both under-weighted whenever it is trained (mean weight 0.061,

below the equal share in 14 of 14 subsets) and the hardest held-out generator (coefficient −12.88 AUC points). Which generators are held out explains 79.5% of OOD variance; after removing it, the correlation is 0.007.

Table 15: M7b summary.

Group	Subsets	Mean $G_{\text{eff}}/\lvert S\rvert$	Mean OOD AUC
All	25	0.824	86.77
NT in training	14	0.696	92.95
NT held out	11	0.987	78.90

Supported: NT is under-weighted in pooled LN tuning; NT as unseen is much harder than DF/FS. Not supported: $G_{\text{eff}}/\lvert S\rvert$ predicts OOD once held-out identity is accounted for.

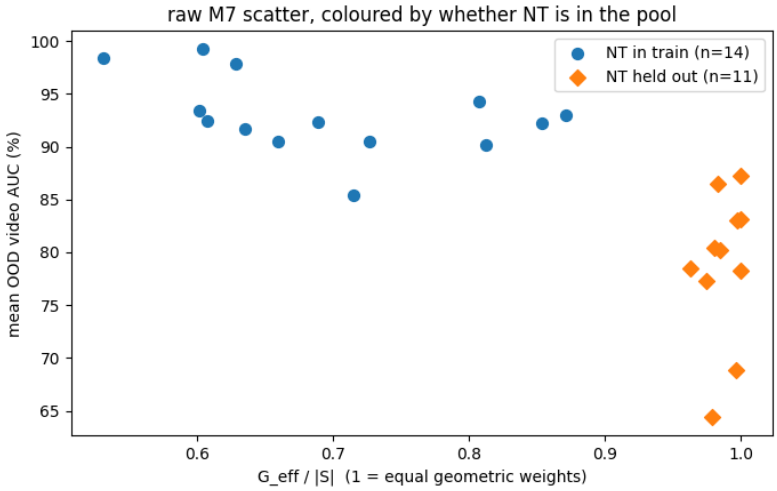


Figure 6: M7b: OOD AUC against $G_{\text{eff}}/\lvert S\rvert$ for the 25 subsets, coloured by whether NT is in the training pool. The two groups occupy different corners, which produces the negative raw correlation.

9.8 M8b — cross-dataset DFD (completed)

Table 16: DFD (363 real / 3,068 fake videos). 95% intervals from an actor-cluster bootstrap. All models: 104,450 deployed parameters.

Model	Video AUC	95% CI	Frame AUC	Video AP	EER	Real / fake acc.
LNCLIP-DF reproduction (M1, ep. 10)	95.88	[94.33, 97.37]	91.44	99.50	11.34	85.95 / 90.42
Pooled, $\lambda = \infty$	96.05	[94.38, 97.65]	92.23	99.52	10.45	83.20 / 92.89
GBLA $\lambda = 1$	96.40	[94.85, 98.01]	92.93	99.57	9.93	85.40 / 92.05
GBLA $\lambda = 0.1$	96.70	[95.20, 98.21]	**93.12**	99.61	9.64	90.63 / 89.83
GBLA $\lambda = 0.01$	**96.71**	[95.30, 98.21]	92.86	**99.61**	**9.06**	92.56 / 89.28
GBLA $\lambda = 0.001$	95.33	[93.78, 96.93]	90.58	99.44	11.87	97.52 / 80.15
GBLA $\lambda = 0$	94.43	[92.74, 96.24]	89.05	99.33	12.93	97.80 / 76.73
Uniform merge	92.68	[90.61, 94.77]	86.65	99.11	15.15	94.77 / 76.73
Fisher merge	94.95	[93.18, 96.80]	89.77	99.40	11.86	72.45 / 92.73
GenD CLIP (paper, 5 seeds)	96.40	–	–	–	–	–

Paired differences (GBLA $\lambda = 0.01$ minus other): vs M1 $+0.82$ $[+0.02, +1.73]$, $P(\Delta > 0) = 0.978$; vs pooled $+0.66$ $[-0.00, +1.40]$; vs $\lambda = 0.1$ $+0.01$ $[-0.29, +0.36]$; vs Fisher $+1.76$ $[+0.74, +2.96]$. Without the three two-person scenes, GBLA − M1 $= +0.87$ $[+0.38, +1.41]$; on single-face frames 97.07 vs 96.44. The ranking over λ on DFD follows the ranking on FF++ FaceShifter. The difference to the paper's 96.4 (5-seed mean, different crops) is not a controlled comparison.

9.9 M9b — cross-dataset UADFV (completed)

Table 17: UADFV (49 real / 49 fake videos). 95% intervals from a pair-cluster bootstrap.

Model	Video AUC	95% CI	Frame AUC	Real / fake acc.	Pairs correct
LNCLIP-DF reproduction (M1, ep. 10)	99.25	[97.46, 100.00]	97.86	95.92 / 97.96	48/49
Pooled, $\lambda = \infty$	99.38	[98.08, 100.00]	97.96	93.88 / 95.92	48/49
GBLA $\lambda = 1$	99.33	[98.00, 100.00]	98.01	85.71 / 97.96	48/49
GBLA $\lambda = 0.1$	99.50	[98.33, 100.00]	98.01	93.88 / 95.92	48/49
GBLA $\lambda = 0.01$	99.63	[98.83, 100.00]	97.97	89.80 / 100.00	**49/49**
GBLA $\lambda = 0.001$	**99.71**	[98.92, 100.00]	**98.04**	100.00 / 95.92	48/49
GBLA $\lambda = 0$	99.58	[98.50, 100.00]	97.81	100.00 / 95.92	48/49
Uniform merge	99.13	[97.25, 100.00]	97.40	97.96 / 95.92	48/49
Fisher merge	98.83	[96.25, 100.00]	97.16	79.59 / 97.96	48/49
GenD CLIP (paper, 5 seeds)	99.20	–	–	–	–

Every paired difference against GBLA $\lambda = 0.01$ has a 95% interval containing 0 (e.g. vs M1 $+0.38$ $[-0.50, +1.92]$). Video EER is 4.08 for every model (one video of 49). UADFV is at ceiling and cannot rank these models; the rank agreement between DFD and UADFV is Spearman 0.38 over the nine models and -0.14 over the six λ values.

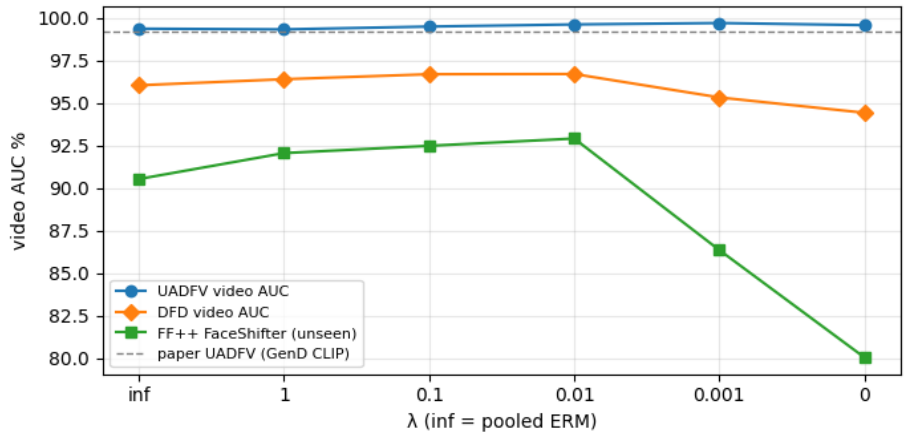


Figure 7: Video AUC against λ on FF++ FaceShifter (unseen generator), DFD and UADFV, with the paper's UADFV value (dashed).

9.10 Verification of the supplied reference values

Table 18: Reference values from earlier discussions, checked against all project files.

Metric	Supplied	Evidence in this project	Status
Total parameters	303,966,208	M0 6.1: ViT-L/14 visual tower *including* the unused final projection (786,432). The detector actually trained has 303,181,826 (projection removed, head added).	verified (different meaning)
Trainable parameters	786,432	In M0/M1 this number is the *removed* CLIP projection, never trained. Every trained model here has 104,450 trainable. 786,432 also equals rank-8 LoRA on query and value of 24 blocks (eq. 10), but no such model exists here.	**unverified**
Trainable %	0.2587%	786,432/303,966,208 = 0.2587% (arithmetic correct). Here: 104,450/303,181,826 = 0.0345%.	**unverified**
LoRA targets	`q_proj`, `v_proj`	No LoRA code; these are Hugging Face module names, the project uses `open_clip`.	**unverified** (not implemented)
LoRA tensors	96	$24 \times 2 \times 2 = 96$ (arithmetic consistent); no file.	**unverified**
Hidden dimension	1024	M0 (width 1,024), M1 counts.	verified
Transformer layers	24	50 LNs and MLP/attention counts imply 24.	verified
Attention heads	16	Standard ViT-L/14 config; not printed.	consistent, not printed
Image resolution	224 × 224	M0 `IMAGE_SIZE=224`; positional count implies 257 tokens.	verified
Patch size	14	Patch embedding 602,112 = $1024 \cdot 3 \cdot 14^2$.	verified
Classifier	Linear(1024→2)	Head 2,050 parameters.	verified
Codebook utilisation	97/1024 (9.47%); 46/1024 (4.49%)	No codebook in the implementation; arithmetic of the percentages is correct.	**unverified**
Best epoch	2/10	No run selects a best epoch by accuracy; M1 best FF++-val epoch is 7/20.	**unverified**
Train acc. / val acc.	98.38% / 97.56%	Per-epoch accuracy is not logged in any notebook.	**unverified**
Val loss	0.134335	Validation loss is not logged in any notebook.	**unverified**
Earlier val acc.	97.125%	Not found.	**unverified**

Interpretation. The unverified values describe a different model: a 303,179,776-parameter CLIP ViT-L/14 vision encoder (the size of this project's frozen tower without head and projection, and of the Hugging Face vision model) with rank-8 LoRA on query and value (303,179,776 + 786,432 = 303,966,208), a codebook and an accuracy-based training log. Neither that model nor its outputs exist in this repository or in the project record. They are kept out of all result tables.

9.11 Summary tables

Table 19: Summary Table 2 — parameter efficiency (all with the frozen CLIP ViT-L/14). Reduction is relative to fine-tuning all 303,181,826 detector parameters.

Configuration	Total	Trainable	Frozen	Trainable %	Reduction
Full visual tower incl. projection (M0)	303,966,208	0	303,966,208	0	–
Full fine-tuning of the detector (hypothetical)	303,181,826	303,181,826	0	100	1×
LN tuning / GBLA deployed	303,181,826	**104,450**	303,077,376	**0.0345**	**2,903×**
GBLA during training ($G = 4$)	303,599,626	522,250	303,077,376	0.172	581×
Head only (M2b)	303,181,826	2,050	303,179,776	0.000676	147,894×
M2 latent $d = 64$ (failed)	303,181,891	65	303,181,826	0.000021	–
M2b latent LN $d = 4096$ (failed)	303,185,923	6,147	303,179,776	0.002027	–
LoRA $r = 8$, q/v (not implemented)	–	(786,432)	–	–	–

Frozen breakdown (M1): MLP 201,449,472; attention 100,761,600; patch embedding 602,112; class + positional 264,192. GBLA training total counts θ_s and four offsets on top of the frozen tower.

Table 20: Summary Table 4 — every training run and its configuration.

Exp.	Runs	Trained on	Parameterisation	Trainable	Epochs × steps	Outcome
M1	1	real + DF, F2F, FS, NT (mixed batches)	LN + head	104,450	20 × 900	FSh 88.37
M2	5	same	latent LN + head, $d \in \{16, 64, 256, 1024, 4096\}$	$d + 1$	10 × 900	failed
M2b	5	same	head only; latent LN $d \in \{64, 256, 1024, 4096\}$ + head	2,050–6,147	10 × 900	failed
M3	4	one generator + 1/4 reals	LN + head	104,450	10 × 225	dominance measured
M4	6	real + 4 gens, pure blocks	GBLA, 6 values of λ	104,450 deployed	10 × 900	best FSh 92.91
M5	8 new + 2 reused	4 of 5 generators	pooled and GBLA $\lambda = 0.01$	104,450	10 × 900	mean +0.55
M6	1 + merges	FaceShifter only; merges	LN + head; post-hoc merges	104,450	10 × 225	Fisher 85.34
M7	20 new + 5 reused	2- and 3-generator subsets	pooled LN + head	104,450	10 epochs	law not supported
M8b, M9b	0	– (evaluation only)	9 checkpoints	104,450	–	DFD 96.71; UADFV 99.63

Table 21: Summary Table 5 — codebook utilisation.

Experiment	Utilisation
Any experiment of this project	*Not evaluated* — no codebook is implemented
Supplied reference (initial)	97/1024 (9.47%) — **unverified**
Supplied reference (later)	46/1024 (4.49%) — **unverified**

Table 22: Summary Table 8 — best verified result per metric (video AUC % unless noted; single seed).

Metric	Value	Produced by
FF++ seen generators (combined)	98.04	M1 LNCLIP-DF reproduction, epoch 20
FF++ seen, 10-epoch budget	97.60	M1 epoch 10
FF++ FaceShifter (unseen)	**92.91**	M4 GBLA $\lambda = 0.01$
LOGO mean held-out	**88.04**	M5 GBLA $\lambda = 0.01$
LOGO held-out NeuralTextures	64.40	M5 pooled $\lambda = \infty$ (GBLA 62.29)
DFD video AUC	**96.71** [95.30, 98.21]	M8b GBLA $\lambda = 0.01$ ($\lambda = 0.1$: 96.70)
DFD frame AUC / EER	93.12 / 9.06	M8b GBLA $\lambda = 0.1$ / $\lambda = 0.01$
UADFV video AUC	99.71 [98.92, 100.00]	M9b GBLA $\lambda = 0.001$ ($\lambda = 0.01$: 99.63; differences not significant)
UADFV pairs correct	49/49	M9b GBLA $\lambda = 0.01$
Deployed trainable parameters	104,450 (0.0345%)	every model above
Training / validation accuracy, validation loss	*Not recorded*	–

10 Technical Challenges and Observations

Table 23: Problems encountered, cause, action and status.

Problem	What happened and why	Action and lesson	Status
"Scrambled" `frames_all`	M0 inspection showed frames of other videos in a folder. The inspection cell listed files without filtering by video; extraction itself filtered correctly.	M1 pairing test (99–100%) and visual grid confirmed the crops. Lesson: verify data with a test on the data, not a display.	resolved
Box source "frames" for 1,002 box files	Expected "videos" after the inspection scare; 1,002 vs 1,000 originals is unexplained.	Pairing test passed; no effect found.	open (minor)
QuickGELU mismatch	M0 loaded CLIP without QuickGELU (warning).	M1 onward uses `force_quick_gelu=True`; parameter counts unaffected.	resolved
Slow training in M1	17–19 min/epoch; image loading from the network file system.	Crops cached in RAM from M2 (17.3 GB).	resolved
Few CPUs in some sessions	M2b had 1 usable CPU: loading 239 img/s became the bottleneck.	Documented; affects run time only.	environment
M2 gradient check	First finite-difference check disagreed by 8.8%: one unit of z moves θ a long way, so the step was too large.	Check redone along a random direction at several step sizes; chain rule matched to 3.3×10^{-7}. Not a bug.	resolved
M2 learning-rate probe	Chose LR by total loss, dominated by uniformity.	M2b probes on cross-entropy.	resolved
Latent cannot express adaptation	M2/M2b failed at $d \leq 1024$.	Compact latent removed from the method.	accepted negative result
Curvature model	Asymmetric, step-dependent, indefinite Hessians; mismatch with geometric weights.	Curvature theory and τ^2 dropped; geometric G_{eff} kept.	accepted negative result
M4: gradient of θ missing	`set_flat` copies values into modules, so θ is not in the graph.	Read gradient from module parameters (`flat_grad`); found before training.	resolved
M4 single-coordinate FD check	One coordinate's slope too small to measure through ViT-L.	Check along a random direction at several steps.	resolved
M7 missing size-4 run	FaceShifter fold existed only under M4.	Path logic fixed to reuse M4.	resolved
Batching confound	$\lambda = \infty$ already +2.3 on FSh vs M1.	All GBLA claims are made against $\lambda = \infty$, not only M1.	accounted for
G_{eff} law	Raw correlation -0.77 driven by NT composition.	Law not claimed (M7b).	accepted negative result
NT held out	GBLA worse than pooled (62.29 vs 64.40).	Reported; limits the generality of the gain.	open
Two-person scenes in DFD	19.4% of frames have ≥ 2 faces; the untouched face may be cropped.	Sensitivity analyses; ranking unchanged.	mitigated
UADFV ceiling	All models 98.8–99.7, CIs overlap.	UADFV not used to rank models.	open
Dataset downloads	Kaggle/API errors for DFD on C-DAC.	Zip uploaded manually; datasets stored outside the repository.	resolved
Single seed	Every number is one run.	Seeds planned.	open

11 Current Architecture Diagrams

Figure 8 shows the implemented inference path and Figure 9 the GBLA training loop. Tensor shapes are written on the arrows. Orange boxes hold the only trained parameters; blue boxes are frozen.

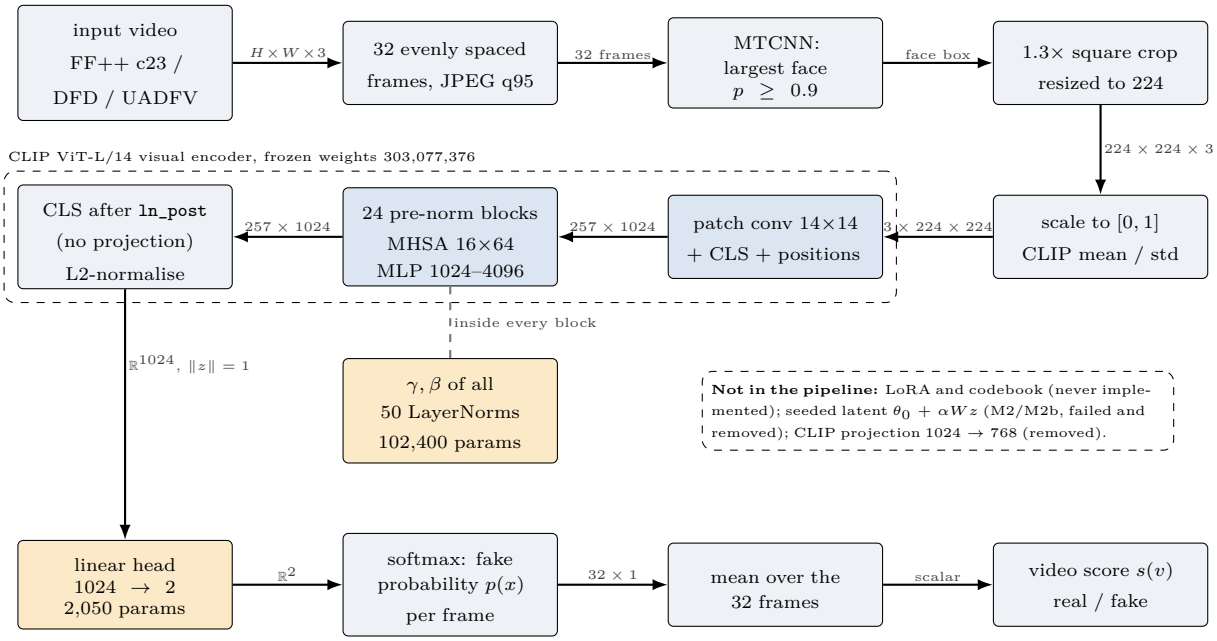


Figure 8: Implemented inference pipeline with tensor shapes. Orange: the only trained parameters, 50 LayerNorms + head = 104,450 numbers (the deployed θ_s); blue: frozen. During training, augmentation is applied between cropping and normalisation.

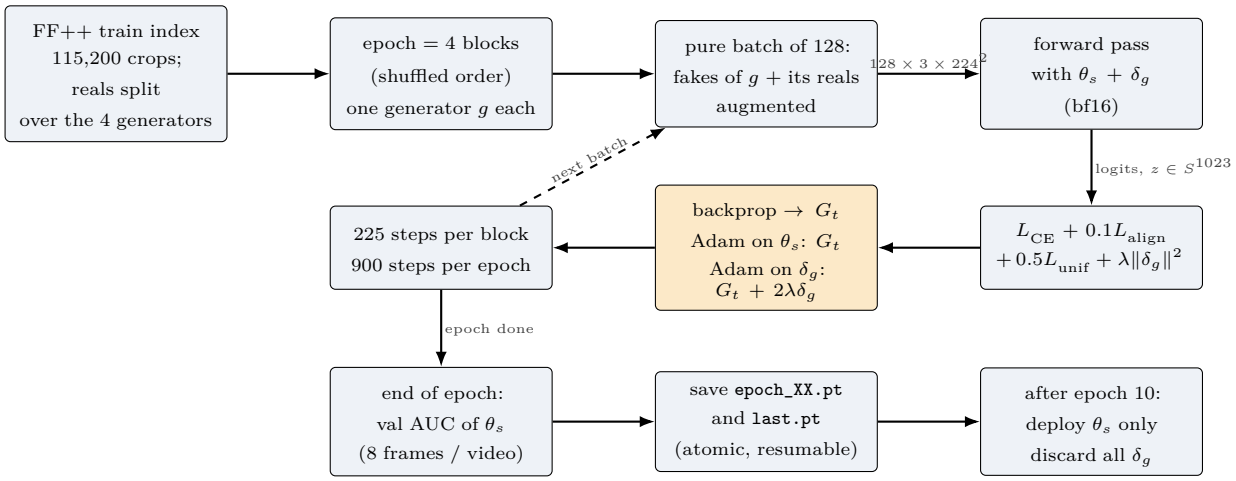


Figure 9: GBLA training workflow (M4 onward). With $\lambda = \infty$ the offsets stay zero, which is pooled training with pure-generator batches; M1 used mixed batches and no offsets. No checkpoint is selected by validation or test performance.

12 Research Contributions

12.1 Implemented and verified

- A resume-safe, leakage-controlled FF++ c23 pipeline with shared real/fake crop boxes and official splits (192,000 crops), and matching DFD and UADFV pipelines.
- A faithful LNCLIP-DF reproduction (within 0.8 AUC points of the paper on six test sets) with exact parameter accounting (104,450 trainable of 303,181,826).
- A measurement of generator dominance in pooled LN tuning: Face2Face carries weight 0.58, NeuralTextures ≈ 0, $G_{\text{eff}}/G = 0.57$; NT is under-weighted in all 14 subsets that contain it.
- GBLA: shared-plus-private optimisation with pure-generator batching, deploying the same 104,450 parameters. Under a fixed 10-epoch budget it improves FaceShifter (92.91 vs 90.53 same-batching pooled vs 88.19 M1), LOGO mean (+0.55) and DFD (+0.82 over M1, interval excluding 0).

12.2 Preliminary findings (need seeds and more data)

- The λ ordering on FF++ FaceShifter carries over to DFD; it does not on UADFV (ceiling).
- GBLA beats a post-hoc Fisher merge of specialists by 2.70 points on LOGO mean, suggesting that joint training with offsets does something a merge does not.
- Pure-generator batching alone changes the generalisation profile (+2.3 FSh, −6.0 NT vs M1).

12.3 Exploratory or theoretical only

- Curvature/random-effects interpretation of λ (failed checks).
- Compact seeded latent (failed), Mapping-Network-style generation, LoRA vs LN comparison, real-only training, codebook/manifold methods: not implemented or abandoned.

12.4 Unresolved questions

- Is the GBLA gain larger than seed variance? (single seed everywhere)
- Why does GBLA hurt held-out NeuralTextures, the hardest generator?
- Does the gain persist on harder benchmarks (Celeb-DF v2, DFDC, FFIW, DeepSpeak)?
- How much of the gain comes from batching vs offsets on other datasets?

Independent assessment. *Parameter efficiency*: demonstrated in the sense of trainable/stored parameters (0.0345%), inherited from LN tuning; GBLA adds none at deployment but 5× at training time; inference cost unchanged. *Detection performance*: in-domain FF++ AUC is near saturation for all methods and does not discriminate. *Generalisation*: measured only by unseen-generator and cross-dataset AUC, never by validation accuracy; gains are modest (0.5–2.4 points) and single-seed.

13 Limitations

- One seed per configuration; no variance estimate over training randomness (bootstrap intervals cover only test sampling).
- MTCNN without landmark alignment at 224 px instead of the paper's RetinaFace alignment at 256 px.
- FF++ validation saturates; no informative validation set is used for model selection (final epoch used everywhere).
- λ was chosen on FaceShifter test AUC in M4 (the only knob tuned this way) and then fixed for M5–M9b; strictly, FaceShifter is not an untouched test set for the choice of λ.
- Only two cross-dataset benchmarks, one of them at ceiling.
- Frame-level model; temporal information and patch tokens are discarded.
- The frozen 303 M backbone dominates inference cost; no compute or latency was measured.

14 Motivation for Architectural Redesign

Worth retaining. The data pipeline and its integrity checks; the frozen-backbone, small-trainable-set paradigm with full parameter accounting; the evaluation protocol (LOGO, cross-dataset, cluster bootstrap, paired differences, load checks).

Needs modification. Batching and balancing across generators (the strongest single lever found, with a cost on NT); validation and model selection (FF++ val is uninformative); number of seeds.

To reconsider. Whether a CLS-only, frame-level representation can carry manipulation evidence for local edits such as NeuralTextures; whether LayerNorm tuning is the right adaptation structure (the LN-vs-LoRA question was never tested).

Missing evaluations. Celeb-DF v2 and other harder datasets; robustness (compression, resizing); compute/latency; multiple seeds; ablation separating batching from offsets on cross-dataset data.

Possible overfitting sources. 1:4 real:fake imbalance; FF++-specific compression (JPEG q95 frames); identity overlap between real and fake in FF++ (controlled by pairing, but identities repeat across frames); selection of λ on FaceShifter.

Questions the redesign must answer. (1) How to improve held-out NT-like local manipulations without losing the FaceShifter gain. (2) How to select hyper-parameters without an unseen test set. (3) Whether gains

survive multiple seeds and harder datasets at the same trainable budget. This report does not propose the new architecture.

15 Future Research Roadmap

All items are future work.

1. Architecture redesign (scope to be agreed with the supervisor).
2. Statistical validation: 3–5 seeds for M1, $\lambda = \infty$ and $\lambda = 0.01$.
3. Cross-dataset testing on Celeb-DF v2 (download notebook prepared outside the repository), DFDC, FFIW, DeepSpeak.
4. Ablations: batching vs offsets; λ chosen without FaceShifter; adapter structure (LN vs LoRA at matched parameter count).
5. Parameter-efficiency comparison including inference compute and memory.
6. Additional manipulation types (DF40-style face swaps, diffusion-based).
7. Robustness evaluation (JPEG, resizing, blur).
8. Reproducibility package: scripts extracted from notebooks, fixed environment file.

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Note. Entries for papers 4–8 were checked against the PDFs in the thesis folder; Tian et al. against arXiv. Standard references (CLIP, FF++, LoRA, etc.) are cited as recorded in the project and should be checked for page numbers before external submission.

A Notation

Table 24: Symbols.

Symbol	Meaning
x, y, g	face crop; label (1 = fake); generator
$\mathcal{G}, G$	training generators; their number (4 in the standard protocol)
P	adapted parameters: 102,400 LayerNorm + 2,050 head = 104,450
$\theta_0, \theta_s, \delta_g$	initial values; shared parameters (deployed); private offset of g (training only)
λ	strength of the offset penalty
v_g	solution of LN tuning on generator g alone (M3, M6)
w_g, G_{eff}	geometric weight of g in the pooled solution; Kish effective number
$z(x)$	L2-normalised CLS feature, $\mathbb{R}^{1024}$
W, z, α, d	(M2/M2b only) seeded random matrix, latent, scale, latent size

B Full M7 subset table

Table 25: M7: geometric G_{eff}, share of the pooled direction explained by the single-generator directions, seen AUC and mean OOD AUC for all 25 subsets.

Subset	$\lvert S\rvert$	$G_{\text{eff}}/\lvert S\rvert$	Expl.	Seen	OOD	Subset	$\lvert S\rvert$	$G_{\text{eff}}/\lvert S\rvert$	Expl.	Seen	OOD
DF+F2F+FS+NT	4	0.659	0.660	96.41	90.53	F2F+FS+FSh	3	0.985	0.726	99.25	80.20
DF+F2F+FS+FSh	4	0.979	0.705	99.15	64.40	F2F+NT+FSh	3	0.531	0.724	95.55	98.37
DF+F2F+NT+FSh	4	0.628	0.702	95.01	97.90	FS+NT+FSh	3	0.689	0.736	90.91	92.29
DF+FS+NT+FSh	4	0.715	0.713	91.94	85.40	DF+F2F	2	0.983	0.736	99.37	86.47
F2F+FS+NT+FSh	4	0.604	0.697	94.42	99.21	DF+FS	2	0.963	0.796	99.86	78.48
DF+F2F+FS	3	0.981	0.739	99.46	80.44	DF+NT	2	0.854	0.680	96.63	92.18
DF+F2F+NT	3	0.607	0.709	96.84	92.43	DF+FSh	2	0.975	0.776	99.79	77.29
DF+F2F+FSh	3	0.998	0.723	99.01	83.02	F2F+FS	2	1.000	0.742	99.35	83.11
DF+FS+NT	3	0.871	0.745	94.04	92.95	F2F+NT	2	0.635	0.725	96.06	91.67
DF+FS+FSh	3	0.996	0.770	99.60	68.89	F2F+FSh	2	1.000	0.734	99.12	87.27
DF+NT+FSh	3	0.727	0.686	92.13	90.48	FS+NT	2	0.807	0.706	95.63	94.24
F2F+FS+NT	3	0.601	0.737	95.36	93.44	FS+FSh	2	0.999	0.765	99.67	78.30
						NT+FSh	2	0.812	0.720	94.03	90.20

C DFD per-scene and sensitivity results

Table 26: DFD sensitivity (video AUC %).

Model	All videos	Without two-person scenes (3,114 videos)	Single-face frames (80.1% of crops)
LNCLIP-DF reproduction (M1)	95.88	97.49	96.44
Pooled, $\lambda = \infty$	96.05	97.65	96.43
GBLA $\lambda = 1$	96.40	98.08	96.85
GBLA $\lambda = 0.1$	96.70	98.39	97.07
GBLA $\lambda = 0.01$	96.71	98.35	97.07
GBLA $\lambda = 0.001$	95.33	96.41	95.74
GBLA $\lambda = 0$	94.43	95.45	94.74
Uniform merge	92.68	93.87	92.84
Fisher merge	94.95	96.36	95.19

The two-person scenes are the hardest for every model (e.g. `meeting_serious`: M1 60.85, pooled 61.79, GBLA $\lambda = 0.01$ 72.17).

D Parameter arithmetic

- Patch embedding $1024 \cdot 3 \cdot 14 \cdot 14 = 602{,}112$; class + positional $1024 + 257 \cdot 1024 = 264{,}192$.
- Attention per block $3072 \cdot 1024 + 3072 + 1024^2 + 1024 = 4{,}198{,}400$; $\times 24 = 100{,}761{,}600$.
- MLP per block $4096 \cdot 1024 + 4096 + 1024 \cdot 4096 + 1024 = 8{,}393{,}728$; $\times 24 = 201{,}449{,}472$.
- LayerNorms $50 \times 2 \times 1024 = 102{,}400$; head $2 \cdot 1024 + 2 = 2{,}050$.
- Detector total $201{,}449{,}472 + 100{,}761{,}600 + 602{,}112 + 264{,}192 + 102{,}400 + 2{,}050 = 303{,}181{,}826$.
- Tower without head and projection 303,179,776; with projection $1024 \times 768 = 786{,}432$: 303,966,208.
- GBLA training store $5 \times 104{,}450 = 522{,}250$.

E Git history

Table 27: Repository commits (oldest first).

Commit	Date	Message (abridged)
2572354	2026-10-05	Initial code added
83cd5dc	2026-10-08	M0 data pre processing pipeline done
3e62dfe	2026-10-13	reproduce LNCLIP-DF on FF++ c23 (frozen CLIP ViT-L/14, train only …)
dfc41db	2026-10-14	generate the LayerNorm + head parameters as theta = theta0 + alpha*W*z … sweep latent size d (16–4096)
6549a0e	2026-10-15	Revised m2 plans in m2b
9f1d12d	2026-10-17	E1 go/no-go on pooled-training generator dominance
0ef0d1f	2026-10-18	GBLA shared LayerNorm+head parameters with private offsets
0731e63	2026-10-19	leave-one-generator-out for GBLA at lambda 0.01
04aa840	2026-10-19	Fisher merging baseline for LOGO comparison
5ef28c8	2026-10-20	G_eff law over 25 generator subsets
7a6758d	2026-10-20	m7 output analysis
439e77a	2026-11-04	Cross Dataset Testing 1 on DFD dataset (M8a preprocessing)
6661977	2026-11-04	Cross Dataset Testing 1 on DFD dataset
f8f6bb7	2026-11-05	Cross Dataset Testing 2 Preprocessing of UADFV dataset
06aedba	2026-11-05	Cross Dataset Testing 2 on UADFV dataset done

F Additional figures

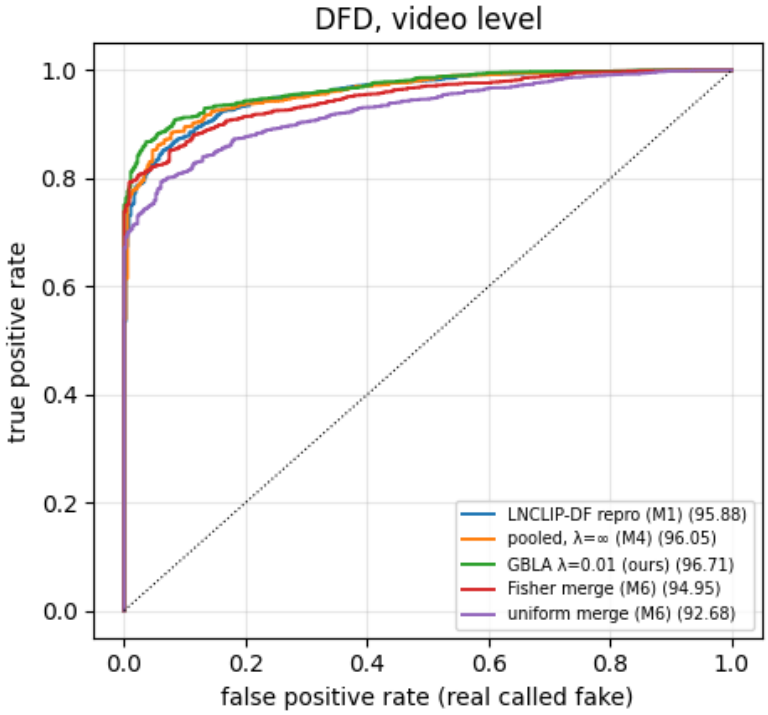


Figure 10: M8b: video-level ROC curves on DFD.